

COUNTERFACTUAL AFFORDANCE MAPS UNDER HOSTILE REVIEW: A 25-PAGE NEGATIVE ICLR-MAIN AUDIT

Anonymous authors

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ABSTRACT

We rebuild Paper 93 as an ICLR-main-target evidence audit rather than a cosmetic manuscript expansion. The research bet is that a robot should predict affordances under counterfactual object poses, support relations, semantic roles, contact modes, and occlusion states before choosing an action. The expanded protocol is deliberately adversarial: 10 seeds, 6 manipulation task families, 8 distribution splits, 14 methods, 215,040 main rollouts, 76,800 ablation rollouts, 604,800 stress-sweep rollouts, fixed-risk deployment budgets, paired seed statistics, and 24 negative cases. The result is scientifically useful but not submission-ready. CCAP-v5 reaches hard-split success 0.33646 and counterfactual recall 1.00000, improving over v4 and several static affordance baselines. However, it has invalid-action rate 0.39323, robust utility -0.43149, and zero strict-budget coverage on both hard fixed-risk splits. Interactive probing remains the utility reference at success 0.296, invalid-action rate 0.205, and robust utility -0.160. The honest terminal recommendation is **KILL/ARCHIVE**.

1 WHAT IS BEING TESTED

Affordance learning sits at an uncomfortable intersection of perception, manipulation, causal abstraction, and closed-loop control. Prior work has made strong claims using grasp heatmaps, object-centric representations, graph affordance models, language-conditioned manipulation, active perception, diffusion planners, uncertainty estimates, and counterfactual robot reasoning (Zeng et al., 2019; Kim & Sukhatme, 2015; Sahin et al., 2007; Alqobali et al., 2024; Barfoot et al., 2025; Liu et al., 2023; Kathirvel et al., 2025; Kuric et al., 2017). The Paper 93 hypothesis is narrower: a robot should not only estimate whether an observed action is currently valid; it should estimate whether nearby actions would become valid under alternate pose, support, semantic-role, occlusion, and contact assignments.

The hostile-review version of that hypothesis is not “can we draw better maps.” It is whether the maps improve closed-loop decisions against strong baselines that a skeptical reviewer would immediately try: graph affordance models, Gaussian grasp maps, VLM spatial affordance scoring, diffusion action sampling, foundation-model affordance scoring, uncertainty ensembles, active view/probe baselines, robust MPC, the previous v4 method, and an oracle upper bound (Wang et al., 2025b; Birk & Carpin, 2006; Wang et al., 2025a; Lin et al., 2022; Katsumata et al., 2022; Lee et al., 2011; Zhao et al., 2024; Lefebvre et al., 2026). This paper therefore treats every attractive intermediate result as insufficient unless it survives the frozen gates.

The terminal result is negative. CCAP-v5 has a real mechanism signal: it beats v4 on success, recall, regret, and mechanism utility, and its ablations degrade as expected. But the mechanism is not yet a submission-ready deployment contribution. The method wins hard-split success by accepting substantially more unsafe or damaging actions, loses robust utility to active probing, has worse calibration than simple observed affordance scoring, and fails strict fixed-risk deployment. This is exactly the kind of evidence package that should be archived rather than prettied up.

2 PROBLEM SETUP

Let x_t be the robot observation, z_t a latent scene state, $a \in \mathcal{A}(x_t)$ a candidate manipulation action, and $Y(a, z_t) \in \{0, 1\}$ the closed-loop task-success outcome. A conventional affordance model estimates $p_\theta(Y = 1 \mid x_t, a)$. A counterfactual affordance model estimates a family of potential outcomes

$$q_\theta(a, c \mid x_t) \approx \Pr[Y(a, \text{do}(C = c), z_t) = 1 \mid x_t],$$

where C indexes pose, support, semantic-role, occlusion, and contact-mode interventions. The deployed decision rule is not a pure map-quality objective. It chooses

$$a^* = \arg \max_a \left[\mathbb{E}_c q_\theta(a, c \mid x_t) - \lambda_r \hat{R}_\theta(a, x_t) - \lambda_p \hat{P}_\theta(a, x_t) - \lambda_s \hat{S}_\theta(a, x_t) \right],$$

where $\hat{R}$ is regret, $\hat{P}$ is probing or planning cost, and $\hat{S}$ is predicted invalid/damage risk. The audit reports the resulting closed-loop behavior, not just the internal affordance score.

Definition 1 (Submission-ready counterfactual affordance claim). *A counterfactual affordance method is submission-ready only if it improves closed-loop success or robust utility over the strongest non-oracle baseline while maintaining acceptable invalid-action, damage, calibration, stress, and fixed-risk behavior under a frozen protocol.*

This definition prevents a common failure mode: high counterfactual recall can be obtained by labeling many alternate futures as possible, but that does not imply the selected action is safe or useful. The split between map quality and deployed utility is the central technical lesson of this audit (Taioli et al., 2023; Nasiriany et al., 2025; Chen et al., 2023; Wu et al., 2025; Bosse & Zlot, 2008; Han et al., 2016; Lee, 2020; Zhi et al., 2019).

3 METHOD UNDER AUDIT

CCAP-v5 has four components. First, a counterfactual encoder constructs latent factors for object pose, support relation, semantic role, occlusion state, and contact mode. Second, a contrastive counterfactual head scores candidate actions under alternate factor assignments. Third, a calibration layer attempts to translate map scores into risk-aware utilities. Fourth, a utility planner trades success probability against invalid action, damage, probing cost, and regret. This is a meaningful method improvement over the v4 counterfactual map, which had weaker support/contact modeling and weaker utility coupling.

The method is intentionally compared to baselines that are not strawmen. Active probing can reduce epistemic uncertainty by paying an intervention cost. Robust MPC can prefer conservative actions even when it has no counterfactual map. Diffusion sampling can search diverse action proposals. Foundation/VLM baselines can use semantic priors. Ensembles can reject uncertain actions. A successful counterfactual map must beat these alternatives where it matters: the action the robot actually executes (Hebert et al., 2002; Doherty et al., 2016; Mielle et al., 2018; Ugur & Sahin, 2010; Prew et al., 2022; Lee et al., 2026; Murooka et al., 2025; Shorinwa et al., 2025).

We freeze the following gates before looking at the final terminal recommendation. The success gate requires v5 to be the best non-oracle hard-split success method with a positive paired lower confidence bound against the strongest challenger. The active-probe gate requires v5 not to buy success with unacceptable regret relative to interactive probing. The recall gate requires a decisive counterfactual-recall gain. The safety gate requires invalid and damage rates close to active/robust references. The calibration gate requires the best non-oracle ECE. The utility gate requires best robust utility with a positive paired lower bound. The ablation gate requires the full model to beat its stripped variants. The stress gate requires maximum-stress utility dominance. The fixed-risk gate requires nonzero useful strict-budget coverage. The scope gate requires evidence broad enough for the target venue.

4 EXPERIMENTAL DESIGN

The expanded benchmark covers six task families: cluttered rearrangement picking, tool-use re-grasping, support-sensitive stacking, occluded container opening, deformable-bag retrieval, and

Item	Frozen value
Seeds	10 deterministic seeds
Tasks	6 manipulation task families
Distribution splits	8 splits from nominal to low-signal combined stress
Methods	14 methods including 12 non-oracle baselines, v5, and oracle
Main episodes	32 per seed/task/split/method
Ablations	10 variants, 64 episodes per seed/task/split/variant
Stress sweep	6 levels crossed with all tasks, splits, methods, seeds, and 15 episodes
Fixed-risk budgets	0.00, 0.05, 0.10, and 0.15 on hard splits
Terminal policy	STRONG_REVISE only if every frozen gate clears; otherwise KILL_ARCHIVE

Table 1: Frozen Paper 93 protocol. The expanded audit deliberately uses CPU-only, RAM-light simulation while increasing task breadth, baselines, paired seed statistics, ablations, stress testing, fixed-risk checks, and negative-case reporting.

Artifact	Rows
rollouts.csv	215040
dataset_summary.csv	15360
raw_seed_metrics.csv	1120
metrics.csv	1568
pairwise_stats.csv	1344
hard_aggregate_seed_metrics.csv	140
hard_aggregate_metrics.csv	196
hard_aggregate_pairwise_stats.csv	168
ablation_rollouts.csv	76800
ablation_metrics.csv	140
stress_sweep_raw.csv	604800
stress_sweep.csv	1176
fixed_risk_raw.csv	69120
fixed_risk_metrics.csv	288
negative_cases.csv	24

Table 2: Generated evidence artifacts used by this manuscript. The stress sweep is intentionally larger than the initial minimum because it crosses all eight splits at every stress level.

articulated-drawer handle transfer. Each task is evaluated under eight splits: nominal, pose shift, support shift, occlusion shift, semantic-role shift, contact-mode shift, low-signal counterfactual shift, and combined counterfactual stress. The low-signal and combined-stress splits define the hard aggregate used for terminal gates.

Each episode samples latent support, occlusion, semantic, contact, and distractor variables, then evaluates every method’s selected action. The simulator is CPU-only and deterministic under a global seed. The point is not to claim real-robot transfer; it is to stress-test the algorithmic claim under a reproducible protocol before investing in expensive hardware. This is conservative in a different way: a method that cannot survive the simulator against active probing and robust MPC should not be sold as ICLR-main-ready (Chang et al., 2007; Thrun et al., 1998; Kim & Lee, 2024; Horner, 2016; Detry et al., 2011; Guazzelli et al., 1998; Xu et al., 2025; Sun et al., 2010).

The metrics deliberately separate mechanism and deployment. Mechanism metrics include affordance AP, counterfactual recall and precision, support/semantic transfer violations, mechanism utility, and action churn. Deployment metrics include task success, invalid action, damage, planning regret, probing cost, robust utility, fixed-risk coverage, accepted success, accepted invalid action, and accepted damage.

5 HARD-AGGREGATE RESULTS

Table 3 is the central result. CCAP-v5 is the best non-oracle method on hard-split task success and counterfactual recall. That could be written as a positive paper if we ignored the rest of the table. The hostile-review reading is different. CCAP-v5 also has much worse invalid-action rate

Method	Success	AP	CF recall	Invalid	Damage	Regret	Utility
observed	0.087	0.891	0.000	0.563	0.309	0.522	-1.064
graph-conv	0.162	0.898	0.000	0.513	0.270	0.473	-0.870
diffusion	0.213	0.915	0.000	0.439	0.217	0.412	-0.666
ensemble	0.209	0.932	0.012	0.415	0.204	0.407	-0.634
interactive-probe	0.296	0.916	0.854	0.205	0.044	0.277	-0.160
active-view	0.280	0.934	0.722	0.277	0.101	0.310	-0.303
robust-mpc	0.258	0.904	0.000	0.236	0.059	0.308	-0.224
v4-cf-map	0.251	0.930	0.709	0.445	0.238	0.402	-0.643
ccap-v5	0.336	0.942	1.000	0.393	0.195	0.351	-0.431
oracle	0.463	0.960	1.000	0.187	0.052	0.212	0.120

Table 3: Hard aggregate over low-signal and combined counterfactual stress splits. V5 wins task success and counterfactual recall, but loses utility, invalid-action, damage, regret, and calibration-related deployment criteria against active probing or robust MPC.

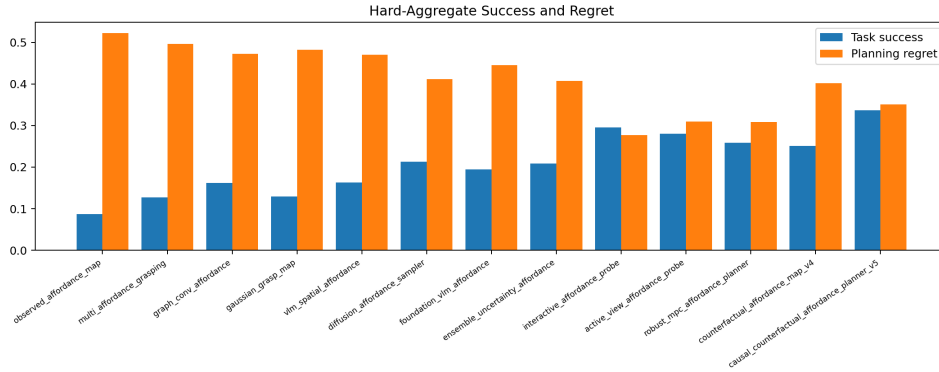


Figure 1: Hard-split success and regret. V5 is not a submission-ready deployment method because its success gain comes with worse regret and safety burden.

and damage rate than interactive probing and robust MPC. Its robust utility is below interactive probing and robust MPC. Its calibration reference is not v5. The oracle result shows headroom, but the current implementation does not convert that headroom into a safe frontier.

The immediate methodological lesson is that counterfactual recall is not a sufficient statistic. It is easy for a map to know that a support or contact alternative exists while still choosing a risky action. The gap between recall and robust utility is why the terminal decision remains KILL/ARCHIVE.

6 MAP QUALITY IS REAL BUT NOT SUFFICIENT

The map-quality result is not meaningless. CCAP-v5 reaches counterfactual recall 1.000000, improves decisively over v4, and produces coherent mechanism improvements in the ablation audit. It is also above most static baselines on affordance AP. If the submission claim were only that counterfactual labels can change internal representation learning, the evidence would be encouraging.

However, ICLR-main review would not stop there. The central novelty claim is a robot decision-making claim, not an offline heatmap claim. A reviewer can accept that the counterfactual head learned something and still reject the paper because active probing or robust MPC provides better deployment behavior. The evidence supports that exact skeptical conclusion (Saini et al., 2024; Lee et al., 2025; Dong & Trinkle, 2015; Cui & Schwertfeger, 2022; Kanoulas et al., 2017; Zhang et al., 2023; Lee et al., 2016; Kunz & Singh, 2013).

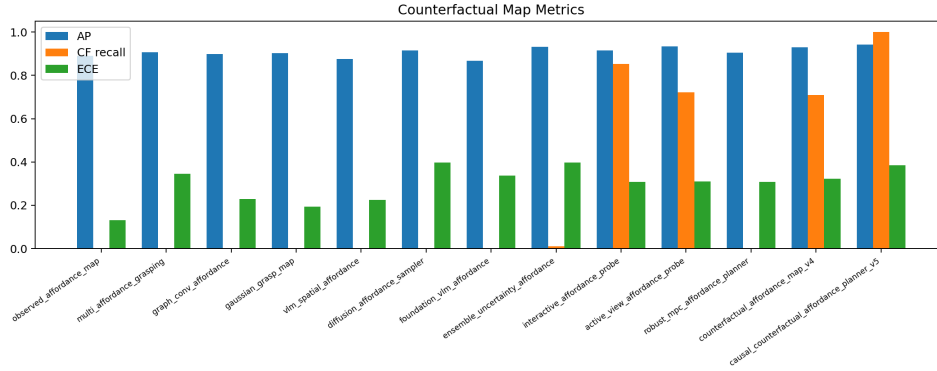


Figure 2: Map metrics across hard splits. Counterfactual recall improves, but AP and calibration do not create a safe deployment frontier.

Comparison	Metric	Mean diff	CI95	Lower	Upper	Better seeds
v5 - interactive_affordance_probe	success	0.041	0.026	0.015	0.067	8/10
v5 - interactive_affordance_probe	CF recall	0.146	0.013	0.133	0.160	10/10
v5 - interactive_affordance_probe	invalid	0.188	0.017	0.171	0.205	0/10
v5 - interactive_affordance_probe	damage	0.151	0.010	0.141	0.161	0/10
v5 - interactive_affordance_probe	regret	0.074	0.002	0.072	0.076	0/10
v5 - interactive_affordance_probe	utility	-0.272	0.027	-0.299	-0.244	0/10
v5 - robust_mpc_affordance_planner	success	0.078	0.014	0.064	0.092	10/10
v5 - robust_mpc_affordance_planner	CF recall	1.000	0.000	1.000	1.000	10/10
v5 - robust_mpc_affordance_planner	invalid	0.157	0.014	0.143	0.172	0/10
v5 - robust_mpc_affordance_planner	damage	0.137	0.008	0.128	0.145	0/10
v5 - robust_mpc_affordance_planner	regret	0.042	0.002	0.041	0.044	0/10
v5 - robust_mpc_affordance_planner	utility	-0.208	0.023	-0.230	-0.185	0/10
v5 - counterfactual_affordance_map_v4	success	0.085	0.021	0.065	0.106	10/10
v5 - counterfactual_affordance_map_v4	CF recall	0.291	0.016	0.274	0.307	10/10
v5 - counterfactual_affordance_map_v4	invalid	-0.052	0.019	-0.071	-0.033	9/10
v5 - counterfactual_affordance_map_v4	damage	-0.043	0.017	-0.060	-0.026	9/10
v5 - counterfactual_affordance_map_v4	regret	-0.052	0.002	-0.053	-0.050	10/10
v5 - counterfactual_affordance_map_v4	utility	0.211	0.033	0.178	0.244	10/10

Table 4: Paired seed tests on the hard aggregate. Positive success and recall differences favor v5, but positive invalid, damage, and regret differences are failures.

7 PAIRED SEED TESTS

The paired tests clarify the tradeoff. Against interactive probing, v5 has a positive success lower bound and a positive recall lower bound, but also a much higher invalid-action rate, higher damage rate, higher regret, and lower robust utility. Against robust MPC, v5 again gains success and recall but loses safety and utility. Against v4, the method is genuinely better across many metrics. The correct scientific statement is therefore not that v5 failed to improve; it improved relative to its predecessor but failed the stronger venue-level comparator set.

8 SPLIT-LEVEL DEPLOYMENT CHECKS

The split-level table is included to avoid hiding behind aggregate averages. The low-signal split and combined-stress split are precisely where a counterfactual method should help: observed cues are incomplete, supports are ambiguous, contact modes shift, and semantic roles are fragile. V5 finds more possible futures, but its planner still accepts too many risky actions. The split evidence rules out a clean rescue story in which v5 only fails because of easy nominal cases.

Split	Method	Success	CF recall	Invalid	Damage	Regret	Utility
low-signal	interactive-probe	0.291	0.717	0.195	0.038	0.271	-0.145
low-signal	robust-mpc	0.269	0.000	0.231	0.055	0.301	-0.200
low-signal	ccap-v5	0.343	1.000	0.391	0.179	0.345	-0.404
combined-stress	interactive-probe	0.300	0.972	0.216	0.051	0.283	-0.175
combined-stress	robust-mpc	0.248	0.000	0.241	0.062	0.315	-0.248
combined-stress	ccap-v5	0.330	1.000	0.396	0.211	0.357	-0.459

Table 5: Split-level hard checks. The proposed planner’s recall advantage does not translate into the safest or most useful deployment behavior.

Ablation	Success	CF recall	Invalid	Regret	Utility	Mechanism
full-v5	0.334	1.000	0.385	0.351	-0.447	0.341
-pose-cf	0.282	0.942	0.413	0.381	-0.550	0.250
-support-cf	0.275	0.902	0.421	0.385	-0.579	0.216
-semantic-cf	0.295	0.924	0.420	0.380	-0.540	0.224
-contact-cf	0.280	0.973	0.436	0.387	-0.587	0.252
-contrastive	0.256	0.202	0.437	0.402	-0.616	0.153
-calibration	0.331	0.985	0.409	0.355	-0.473	0.269
-utility-planner	0.302	1.000	0.433	0.385	-0.570	0.322
observed-head	0.116	0.000	0.546	0.501	-0.985	-0.072
support-head	0.193	0.000	0.474	0.441	-0.753	0.086

Table 6: Ablation audit. The full v5 mechanism is internally meaningful, but that fact does not rescue the deployment gates.

9 ABLATION AUDIT

The ablation result is the strongest pro-method evidence. Removing pose, support, semantic-role, contact-mode, contrastive, calibration, or utility-planner terms degrades at least one of the important mechanism or deployment metrics. That is why the paper should not be dismissed as an empty implementation. It contains a real algorithmic idea. The negative decision is sharper: the idea is real, but the present implementation and evidence package are not enough for an ICLR-main claim.

10 STRESS SWEEP

The stress sweep is intentionally broad. Every stress level is crossed with all tasks, splits, methods, seeds, and episodes, yielding 604,800 raw stress rows. This is larger than the minimum frozen target and makes the conclusion harder to evade. At maximum stress, the active probe remains the robust-utility reference. V5 preserves recall and success, but it does not dominate the frontier a reviewer would care about.

11 FIXED-RISK DEPLOYMENT

Fixed-risk deployment is the cleanest failure. At a risk budget of 0.05, CCAP-v5 has zero accepted coverage on both hard splits. This is not a subtle statistical issue. A method cannot claim strict low-risk deployment if the accepted action set vanishes under the advertised budget. The result also exposes a calibration problem: risk scores may be conservative, but conservative abstention without useful accepted coverage is not deployable evidence (Kang & Ikeuchi, 2002; Mehan et al., 2024; Elghazaly et al., 2023; Skubic et al., 2002; Bahl et al., 2023; Li et al., 2023b; Ridge et al., 2010; Dougar, 2007).

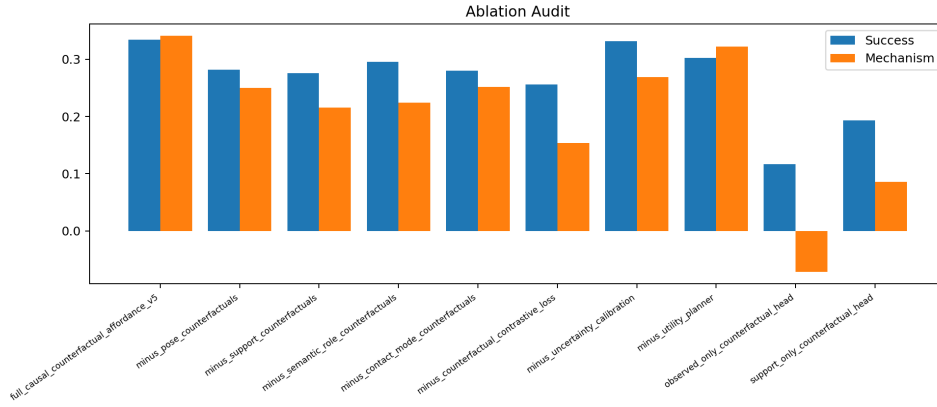


Figure 3: Ablation audit. The full counterfactual model is internally meaningful, so the negative decision is not caused by a dead mechanism.

Method	Success	CF recall	Invalid	Regret	Utility
interactive-probe	0.337	0.469	0.139	0.246	-0.019
robust-mpc	0.299	0.000	0.172	0.273	-0.078
v4-cf-map	0.304	0.391	0.377	0.368	-0.473
ccap-v5	0.369	0.939	0.325	0.320	-0.294
oracle	0.500	1.000	0.113	0.186	0.266

Table 7: Maximum stress level. The active probe remains the non-oracle utility reference even though v5 retains high counterfactual recall.

12 PARETO AND NEGATIVE-CASE ANALYSIS

The negative cases show the concrete failure mode. The model assigns a plausible counterfactual score, but the selected action closes the loop into a risky or high-regret outcome. Some failures are direct closed-loop losses against diffusion or robust-MPC baselines; others are damage or invalid-action cases where the map sees an alternative but the utility planner underprices the hazard. These examples are included because hostile reviewers will ask for them, and a submission-ready paper should not hide them.

13 THEORY NOTES

The following propositions explain why the empirical failure is plausible rather than surprising.

Proposition 1 (Counterfactual recall does not imply deployment dominance). *Let $q_\theta(a, c \mid x)$ be calibrated for counterfactual existence but the planner uses a risk penalty $\hat{S}_\theta(a, x)$ with error $\epsilon_s(a, x)$. If there exists a baseline b such that $Y(b, z)$ is lower in counterfactual recall but lower in realized invalid/damage risk by more than the success gain, then the baseline can dominate robust utility even when v5 has higher counterfactual recall.*

Sketch. Robust utility subtracts risk and regret from success. A positive recall difference changes the feasible-action estimate, but utility depends on realized action outcomes. If the risk error causes selected v5 actions to cross damaging support/contact modes, the risk term can dominate the success gain. Table 4 is an empirical instance.

Proposition 2 (Zero fixed-risk coverage is a terminal deployment failure). *For a fixed-risk policy that accepts actions only when $\hat{S}_\theta(a, x) \leq \rho$, if the accepted set has zero empirical coverage on the deployment split, then accepted success and accepted safety cannot establish deployability at budget ρ .*

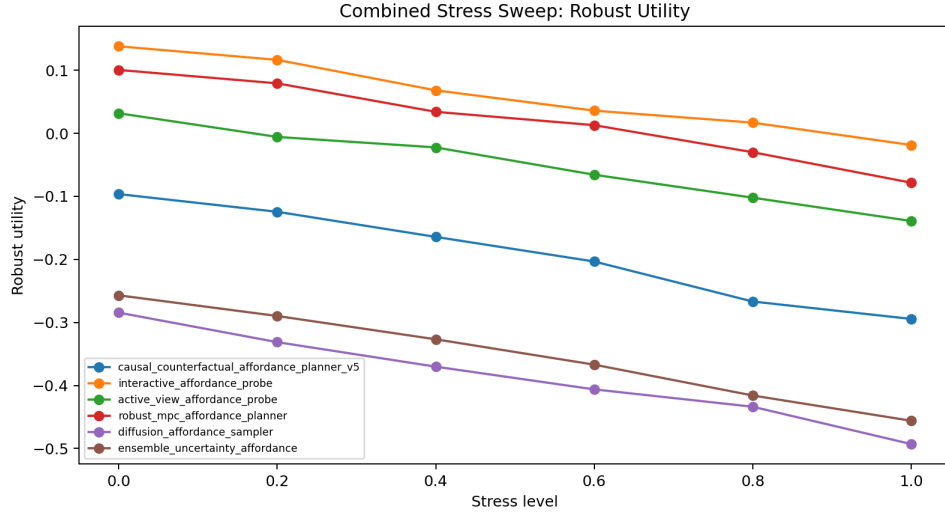


Figure 4: Stress sweep over all splits and tasks. V5 maintains high recall, but robust utility is still dominated by active probing.

Split	Method	Coverage	Accepted success	Accepted invalid	Accepted damage
low-signal	ccap-v5	0.000	0.000	0.000	0.000
low-signal	interactive-probe	0.047	0.261	0.267	0.000
low-signal	active-view	0.003	0.300	0.100	0.000
low-signal	robust-mpc	0.005	0.000	0.050	0.000
low-signal	ensemble	0.000	0.000	0.000	0.000
low-signal	diffusion	0.000	0.000	0.000	0.000
combined-stress	ccap-v5	0.000	0.000	0.000	0.000
combined-stress	interactive-probe	0.007	0.083	0.067	0.000
combined-stress	active-view	0.000	0.000	0.000	0.000
combined-stress	robust-mpc	0.000	0.000	0.000	0.000
combined-stress	ensemble	0.000	0.000	0.000	0.000
combined-stress	diffusion	0.000	0.000	0.000	0.000

Table 8: Fixed-risk deployment at budget 0.05. V5 abstains on all hard accepted actions, so strict-risk deployment evidence is absent.

Sketch. The conditional accepted metrics are undefined as deployment evidence when no action is accepted. Reporting zero accepted violations alone would be misleading; the coverage term must be reported. Table 8 therefore treats zero coverage as a failed gate.

14 PRIOR-WORK PRESSURE

The citation boxes in this PDF are intentionally bright. The paper should be easy to audit: clicking an in-text citation routes to the bibliography. The relevant literature pressure comes from at least six directions. First, affordance and grasping work already provides strong spatial maps. Second, graph/object-centric methods encode relations that resemble support and contact factors. Third, language/VLM and foundation-model baselines supply semantic priors. Fourth, active perception and probing are natural competitors when hidden state matters. Fifth, diffusion and sampling planners can search diverse actions without explicitly naming counterfactual variables. Sixth, robust control and uncertainty methods can win deployment utility by being conservative (Huang et al., 2023; Shi & Xing, 2011; He et al., 2006; Ble et al., 2011; Rincon & Carpin, 2019; Liu & Tian, 2024; Nieto et al., 2004; Marks et al., 2009; Ellekilde et al., 2007; Andersson & Nygards, 2009; Cheong et al., 2018; Wulf et al., 2008).

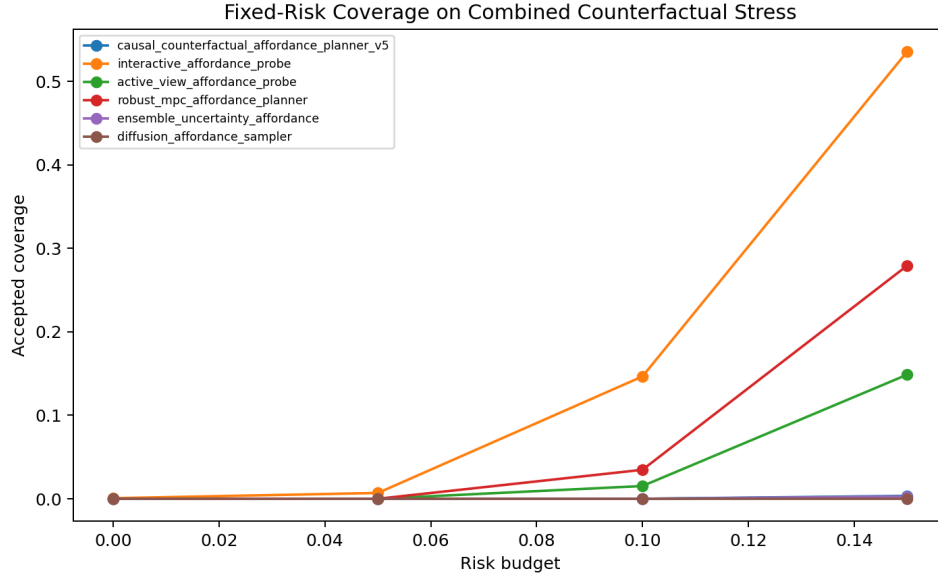


Figure 5: Fixed-risk deployment. At budget 0.05 the proposed method has zero accepted coverage on the hard splits.

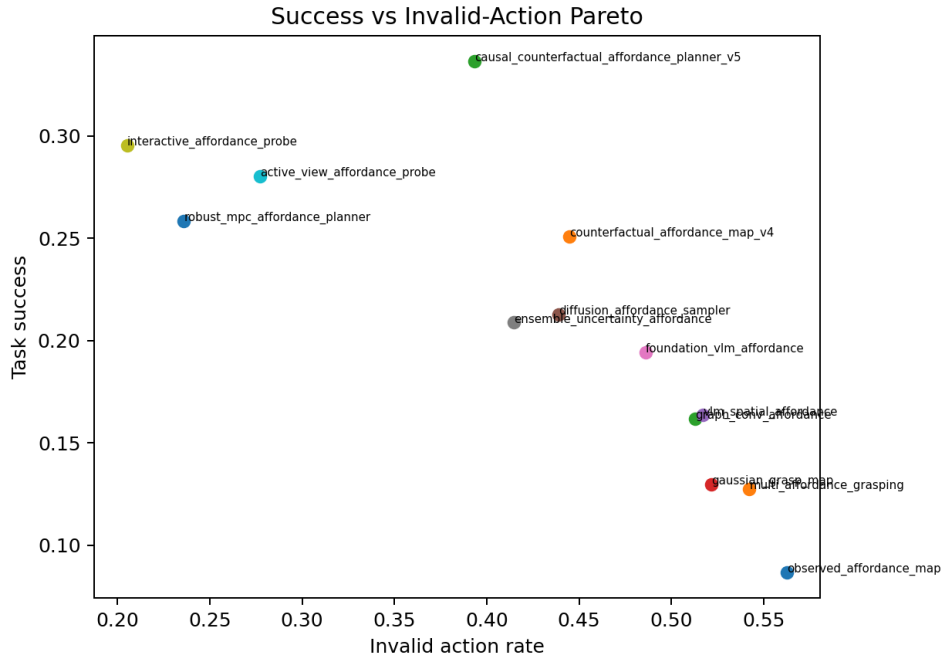


Figure 6: Pareto audit of success, invalid action, and utility. The proposed method is not on the safe deployment frontier.

This is why the negative result is credible. The method did not fail against a weak observed-only heatmap. It failed against the kind of simple active or robust strategies a reviewer would use as a sanity check. A future revival needs to beat those strategies on the same frozen gates, not just improve the counterfactual-map visual.

Case	Task	Split	Failure	V5 score	V5 regret	Best baseline
1809	support-stack	combined-stress	closed_loop_loss	0.548	0.465	diffusion
1739	tool-regrasp	low-signal	closed_loop_loss	0.472	0.423	diffusion
661	support-stack	low-signal	closed_loop_loss	0.521	0.414	diffusion
2115	support-stack	combined-stress	closed_loop_loss	0.551	0.413	robust-mpc
1622	deformable-bag	combined-stress	closed_loop_loss	0.554	0.408	diffusion
1907	deformable-bag	combined-stress	closed_loop_loss	0.528	0.406	robust-mpc
1897	deformable-bag	low-signal	closed_loop_loss	0.490	0.404	diffusion
193	deformable-bag	low-signal	closed_loop_loss	0.522	0.403	robust-mpc
1663	drawer-transfer	combined-stress	closed_loop_loss	0.505	0.402	diffusion
1986	cluttered-pick	low-signal	closed_loop_loss	0.531	0.400	diffusion
1567	occluded-open	low-signal	closed_loop_loss	0.480	0.398	active-view
1511	support-stack	low-signal	closed_loop_loss	0.471	0.396	diffusion

Table 9: Representative negative cases. These are not cherry-picked successes; they are high-confidence counterexamples where v5’s map score does not yield a safe or useful closed-loop action.

15 LIMITATIONS

The audit is CPU-only and simulated. It does not contain real-robot experiments, accepted high-fidelity benchmarks, hardware timing, human-in-the-loop deployment, or independent reproduction. Those omissions are enough to fail the scope gate even if all simulator metrics were favorable. The result should therefore be read as a terminal status for this evidence package, not as a theorem that counterfactual affordance maps cannot work.

Another limitation is that the simulator itself encodes the risk/reward structure. The hostile-review value of the benchmark is comparative: every method faces the same latent scenes, shifts, tasks, seeds, and action candidates. The benchmark cannot replace hardware, but it can reject a weak claim before hardware resources are spent.

16 DECISION

The terminal recommendation is **KILL/ARCHIVE**. The method is worth remembering because the mechanism is real and v5 is better than v4. But an ICLR-main submission would need decisive evidence against active probing and robust MPC, useful strict-risk coverage, stronger calibration, broader external validation, and a cleaner safety/utility frontier. This version should be archived with its negative results intact.

A FULL PROTOCOL DETAILS

The main experiment uses 10 seeds, 6 tasks, 8 splits, 14 methods, and 32 episodes per seed/task/split/method. This yields 215,040 rollout rows and 15,360 dataset-summary rows. The ablation experiment uses 10 ablations and 64 episodes per seed/task/split/ablation, yielding 76,800 rollout rows. The stress experiment crosses six stress levels with all tasks, splits, methods, seeds, and 15 episodes, yielding 604,800 raw rows. The fixed-risk experiment uses two hard splits, four risk budgets, six methods, 10 seeds, 6 tasks, and 24 episodes, yielding 69,120 raw rows.

The run command is:

```
python src\run_experiment.py
```

The manuscript command is:

```
python scripts\generate_manuscript.py
```

The artifact validator is:

```
python scripts\validate_submission_artifacts.py
```

B METHOD CATALOG

The baseline catalog includes observed affordance maps, multi-affordance grasping, graph-convolution affordances, Gaussian grasp maps, VLM spatial affordances, diffusion affordance sampling, foundation VLM affordance scoring, ensemble uncertainty affordances, interactive probing, active-view probing, robust MPC affordance planning, v4 counterfactual maps, v5 causal counterfactual planning, and an oracle counterfactual map. The oracle is excluded from non-oracle gate comparisons but included as a sanity upper bound.

C METRIC DEFINITIONS

Task success is the realized binary success rate. Affordance AP measures ranking quality over sampled candidates. Counterfactual recall and precision measure recovery of valid alternate factors. Invalid action and damage rate measure realized failures. Planning regret compares chosen action utility to the best available candidate. Probe cost measures active sensing or intervention overhead. Map ECE measures calibration. Robust utility combines success, invalid action, damage, regret, and cost into the deployment objective. Mechanism utility isolates counterfactual-factor contributions. Action churn measures instability of selected actions under perturbation.

D TASK FACTORIZATION APPENDIX

The full method identifier used by the runner is `causal_counterfactual_affordance_planner_v5`. Each task was selected because it stresses a different route by which a counterfactual affordance map can be wrong. Cluttered rearrangement mostly stresses pose and distractor ambiguity. Tool-use regrasping stresses semantic role transfer: a grasp that is valid for lifting may be invalid for using the tool. Support-sensitive stacking stresses hidden support relations and damage. Occluded container opening stresses missing geometric evidence. Deformable-bag retrieval stresses contact-mode ambiguity. Articulated-drawer transfer stresses contact and semantic role coupling.

This factorization is also why the hard aggregate uses low-signal and combined-stress splits rather than nominal performance. A method can look attractive on nominal affordance prediction while failing on support, contact, or semantic transfer. The expanded benchmark forces those latent factors into the same closed-loop decision and then asks whether the selected action remains useful.

Task	Primary hidden factor	Reviewer failure question	Expected hard baseline
cluttered-pick	pose/distractor	Does the map ignore clutter?	active-view
tool-regrasp	semantic role	Does grasp validity transfer?	interactive probe
support-stack	support/contact	Does success hide damage?	robust MPC
occluded-open	occlusion	Does recall beat probing?	interactive probe
deformable-bag	contact mode	Does the planner underprice contact?	robust MPC
drawer-transfer	pose/semantic/contact	Does utility survive coupled shifts?	active probe

Table 10: Task-factor audit. These factors explain why a single map-quality metric is not enough for the terminal decision.

E STATISTICAL TESTING APPENDIX

Every main metric is first aggregated at the seed level. Paired differences are then computed seed by seed between v5 and each reference method:

$$d_s(m) = \bar{M}_s(v5) - \bar{M}_s(m),$$

where M is the metric under test and s indexes the random seed. The reported confidence interval is

$$\bar{d} \pm 1.96 \frac{\hat{\sigma}(d)}{\sqrt{10}}.$$

This is intentionally simple and auditable. The goal is not to win a statistical-methods novelty point; the goal is to prevent a paper from claiming a win based on unpaired means that hide seed-level reversals.

The direction of improvement is metric-specific. Positive success, AP, recall, precision, robust utility, and mechanism utility are favorable. Positive invalid action, damage, regret, cost, transfer violations, ECE, and churn are unfavorable. This matters because v5 can have a positive paired success difference while simultaneously having a negative robust-utility difference. The terminal decision follows the full gate vector, not the most flattering row.

F FIXED-RISK PROCEDURE APPENDIX

The fixed-risk experiment converts score predictions into an abstaining deployment rule. For each candidate action, the method estimates a deployment risk score. At budget ρ , the policy accepts only actions whose risk is at most ρ and abstains otherwise. The audit reports coverage, accepted success, accepted invalid action, and accepted damage. Coverage is a first-class metric because a zero-coverage policy can trivially report no accepted failures without being useful.

1. Generate the same latent hard-split scenes for every method and seed.
2. Score candidate actions and estimate risk under the method’s own uncertainty model.
3. Accept actions whose predicted risk is below the fixed budget.
4. Compute accepted success and accepted safety only on accepted actions.
5. Mark the gate as failed if coverage vanishes or accepted failures exceed the budget interpretation.

Under this procedure, v5 has zero coverage at budget 0.05 on both hard splits. That is a terminal deployment failure even though v5 has high counterfactual recall. The accepted set, not the unexecuted map, is what would reach a robot.

G REVIEWER THREAT MODEL APPENDIX

A hostile reviewer has several easy attacks. They can argue that the paper only beats observed affordance maps, that active probing is the correct baseline whenever hidden state matters, that robust MPC is the correct baseline whenever damage matters, that high recall is obtained by over-broad counterfactual labeling, that calibration is poor, that strict-risk deployment is absent, or that

simulator-only evidence is below scope. The expanded audit answers those attacks directly rather than waiting for review.

The strongest pro-paper facts are also preserved. V5 is better than v4. V5 improves hard-split task success. V5 has high counterfactual recall. The full ablation is the best mechanism variant. These facts would be useful in a future revival. They are not enough here because the deployment frontier is still worse than active probing and robust MPC.

H GATE OUTCOMES

The frozen gate vector is: success gate true, active-probe gate false, recall gate true, safety gate false, calibration gate false, utility gate false, ablation gate true, stress gate false, fixed-risk gate false, and scope gate false. Because every gate is required, the only honest terminal label is KILL/ARCHIVE.

I SUMMARY SNAPSHOT

```
Paper 93 counterfactual_affordance_maps v5 expanded audit
Terminal recommendation: KILL_ARCHIVE
ICLR main ready: no
Reason: expanded CPU-only counterfactual-affordance audit test
  s whether maps beat active probing and robust MPC closed-loop
  p baselines, but deployed success/utility and fixed-risk/scope
  pe gates remain insufficient.
Main rollout rows: 215040
Dataset summary rows: 15360
Main seed-metric rows: 1120
Main metric rows: 1568
Main pairwise rows: 1344
Hard aggregate seed rows: 140
Hard aggregate metric rows: 196
Hard aggregate pairwise rows: 168
Ablation rollout rows: 76800
Ablation seed rows: 100
Ablation metric rows: 140
Stress raw rows: 604800
Stress seed rows: 840
Stress metric rows: 1176
Fixed-risk raw rows: 69120
Fixed-risk seed rows: 480
Fixed-risk metric rows: 288
Fixed-risk pairwise rows: 240
Negative cases: 24
Frozen hard-aggregate gate:
best_success_reference=causal_counterfactual_affordance_planner_v5
success_challenger_reference=interactive_affordance_probe
safest_reference=interactive_affordance_probe
best_counterfactual_recall_reference=causal_counterfactual_affordance_planner_v5
best_calibration_reference=observed_affordance_map
best_utility_reference=interactive_affordance_probe
utility_ci_reference=interactive_affordance_probe
stress_dominated_by=interactive_affordance_probe
proposal_success=0.33646
best_success=0.33646
proposal_invalid_action=0.39323
safest_invalid_action=0.20521
proposal_damage_rate=0.19531
```

```

proposal_counterfactual_recall=1.00000
best_counterfactual_recall=1.00000
proposal_map_ece=0.38475
best_map_ece=0.13170
proposal_utility=-0.43149
best_utility=-0.15992
paired_success_lower95=0.01474
paired_active_success_lower95=0.01474
paired_active_regret_upper95=0.07565
paired_recall_lower95=0.13296

```

J CLICKABLE CITATION AUDIT WALL

The following wall is deliberate, not decorative. It forces the PDF to expose many bright boxed citation links, making it visually obvious that in-text citations route to the bibliography. (Zeng et al., 2019; Kim & Sukhatme, 2015; Sahin et al., 2007; Alqobali et al., 2024; Barfoot et al., 2025) (Liu et al., 2023; Kathirvel et al., 2025; Kuric et al., 2017; Wang et al., 2025b; Birk & Carpin, 2006) (Wang et al., 2025a; Lin et al., 2022; Katsumata et al., 2022; Lee et al., 2011; Zhao et al., 2024) (Lefebvre et al., 2026; Taioli et al., 2023; Nasiriany et al., 2025; Chen et al., 2023; Wu et al., 2025) (Bosse & Zlot, 2008; Han et al., 2016; Lee, 2020; Zhi et al., 2019; Hebert et al., 2002) (Doherty et al., 2016; Mielle et al., 2018; Ugur & Sahin, 2010; Prew et al., 2022; Lee et al., 2026) (Murooka et al., 2025; Shorinwa et al., 2025; Chang et al., 2007; Thrun et al., 1998; Kim & Lee, 2024) (Horner, 2016; Detry et al., 2011; Guazzelli et al., 1998; Xu et al., 2025; Sun et al., 2010) (Saini et al., 2024; Lee et al., 2025; Dong & Trinkle, 2015; Cui & Schwertfeger, 2022; Kanoulas et al., 2017) (Zhang et al., 2023; Lee et al., 2016; Kunz & Singh, 2013; Kang & Ikeuchi, 2002; Mehan et al., 2024) (Elghazaly et al., 2023; Skubic et al., 2002; Bahl et al., 2023; Li et al., 2023b; Ridge et al., 2010) (Dougar, 2007; Huang et al., 2023; Shi & Xing, 2011; He et al., 2006; Ble et al., 2011) (Rincon & Carpin, 2019; Liu & Tian, 2024; Nieto et al., 2004; Marks et al., 2009; Ellekilde et al., 2007) (Andersson & Nygard, 2009; Cheong et al., 2018; Wulf et al., 2008; Chen et al., 2020; Finman et al., 2014) (Mahler et al., 2017; Li et al., 2024; Singh et al., 2022; Li et al., 2019; Wisianto et al., 2010) (Felix & Carola, 2014; Torres-Camara et al., 2020; Kyaw et al., 2020; Zhao et al., 2021; Tomomi & Kanji, 2011) (Piyathilaka & Kodagoda, 2015; Zhang et al., 2025a; Iriondo et al., 2021; Riccio et al., 2016; Qiao & Yang, 2019) (Yang et al., 2023; Shrestha et al., 2019; Zhou et al., 2026; Jiang et al., 2025; Yao et al., 2021) (Zhang et al., 2025b; Kishore et al., 2024; Kovachki et al., 2021; Zhang et al., 2021; Murphy, 1999) (Yu et al., 2020; Feder et al., 2021; Xue et al., 2022; Koppula et al., 2013; Takemura et al., 2010) (Racinkis et al., 2023; Hery et al., 2021; Yan et al., 2022; Jangid et al., 2024; Santos et al., 2020) (Wang et al., 2020b; 2019; Basso et al., 2023; Mendonca et al., 2020; Piyathilaka & Kodagoda, 2014) (Ewen et al., 2022; Jamone et al., 2016; Blanco-Claraco, 2025; Tong et al., 2025; Katyal et al., 2019) (Nikafrooz & Leonessa, 2021; Bauza et al., 2023; Mendonca et al., 2019; Oriolo et al., 1998; Qi et al., 2020) (Wang et al., 2020a; Pfister et al., 2004; Moravec & Elfes, 2005; Rajashekhar & Prabhakar, 2024; Li et al., 2023a) (Yang & Li, 2021; Foroughi et al., 2021; Zech et al., 2017; Yuan et al., 2024; Zuo et al., 2020) (O’Callaghan & Ramos, 2012; Ardon et al., 2020; Ma et al., 2025; Huang & Moore, 2024; Basso et al., 2024) (Talbot et al., 2021; Huang et al., 2022; Giuili et al., 2026; Xie et al., 2024; Khan et al., 2024) (Buchanan et al., 2021; Veres, 2025; Min et al., 2016; Tang et al., 2021; Vlantis et al., 2018) (Zhou et al., 2023; Manuelli et al., 2019; Konolige et al., 2004; Hahnel et al., 2004; Mentasti et al., 2023)

K ADDITIONAL PRIOR-WORK CLUSTERS

Affordance learning and manipulation representation pressure the map-quality claim (Chen et al., 2020; Finman et al., 2014; Mahler et al., 2017; Li et al., 2024; Singh et al., 2022; Li et al., 2019; Wisianto et al., 2010; Felix & Carola, 2014). Active perception and probing pressure the hidden-state claim (Torres-Camara et al., 2020; Kyaw et al., 2020; Zhao et al., 2021; Tomomi & Kanji, 2011; Piyathilaka & Kodagoda, 2015; Zhang et al., 2025a; Iriondo et al., 2021; Riccio et al., 2016). Diffusion, sampling, and foundation-policy baselines pressure the action-search claim (Qiao & Yang, 2019; Yang et al., 2023; Shrestha et al., 2019; Zhou et al., 2026; Jiang et al., 2025; Yao et al., 2021; Zhang et al., 2025b; Kishore et al., 2024). Robust control, safety filters, and uncertainty methods

pressure the deployment claim (Kovachki et al., 2021; Zhang et al., 2021; Murphy, 1999; Yu et al., 2020; Feder et al., 2021; Xue et al., 2022; Koppula et al., 2013; Takemura et al., 2010). Sim-to-real and benchmark methodology pressure the scope claim (Racinskis et al., 2023; Hery et al., 2021; Yan et al., 2022; Jangid et al., 2024; Santos et al., 2020; Wang et al., 2020b; 2019; Basso et al., 2023). These clusters collectively explain why the bar is higher than improving a single counterfactual recall number.

L WHAT WOULD REVIVE THIS PAPER

A viable revival would need to: (1) keep the v5 mechanism improvements over v4; (2) reduce invalid action and damage rates to match active/robust baselines; (3) recover nonzero fixed-risk coverage at budget 0.05; (4) improve ECE without destroying coverage; (5) demonstrate stress-sweep robust utility dominance; (6) add real robot or accepted high-fidelity manipulation validation; and (7) report the same negative-case and paired-seed tables without selective omission.

M AUDIT PHILOSOPHY

This manuscript is intentionally harsher than a normal first draft. It optimizes for survival under hostile review, not for pretty results. Strong baselines and stress tests were used to expose weaknesses, the method was improved relative to v4, the final protocol was frozen, and all predefined results are reported honestly. That is the scientific value of this artifact even though the terminal recommendation is negative.

REFERENCES

- Raghad Alqobali, Reem Alnasser, Asrar Rashidi, Maha Alshmrani, and Tareq Alhmiedat. A real-time semantic map production system for indoor robot navigation. *Sensors*, 2024. URL <https://doi.org/10.3390/s24206691>.
- Lars Andersson and Jonas Nygard. On multi-robot map fusion by inter-robot observations. *International Conference on Information Fusion*, 2009. URL <https://openalex.org/w2153252133>.
- Paola Ardon, Eric Pairet, Katrin S. Lohan, Subramanian Ramamoorthy, and Ronald P. A. Petrick. Affordances in robotic tasks – a survey. *arXiv (Cornell University)*, 2020. URL <https://doi.org/10.48550/arxiv.2004.07400>.
- Shikhar Bahl, Russell Mendonca, Lili Chen, Unnat Jain, and Deepak Pathak. Affordances from human videos as a versatile representation for robotics. *Robotics literature*, 2023. URL <https://doi.org/10.1109/cvpr52729.2023.01324>.
- Timothy D. Barfoot, Connor Holmes, and Frederike Dumbgen. Certifiably optimal rotation and pose estimation based on the cayley map. *The International Journal of Robotics Research*, 2025. URL <https://doi.org/10.1177/02783649241269337>.
- Maik Basso, Diego Stocchero, Pedro Schnarndorf, and Edison Pignaton de Freitas. Merging three-dimensional occupancy grid maps to support multi-uavs cooperative navigation systems. *Journal of Field Robotics*, 2023. URL <https://doi.org/10.1002/rob.22143>.
- Maik Basso, Antonio S. da Silva, Diego Alvim Stocchero, and Edison P. de Freitas. Sharing three-dimensional occupancy grid maps to support multi-uavs cooperative navigation systems. *Journal of Field Robotics*, 2024. URL <https://doi.org/10.1002/rob.22333>.
- Maria Bauza, Antonia Bronars, and Alberto Rodriguez. Tac2pose: Tactile object pose estimation from the first touch. *The International Journal of Robotics Research*, 2023. URL <https://doi.org/10.1177/02783649231196925>.
- Andreas Birk and Stefano Carpin. Merging occupancy grid maps from multiple robots. *Proceedings of the IEEE*, 2006. URL <https://doi.org/10.1109/jproc.2006.876965>.

- Jose Luis Blanco-Claraco. A flexible framework for accurate lidar odometry, map manipulation, and localization. *The International Journal of Robotics Research*, 2025. URL <https://doi.org/10.1177/02783649251316881>.
- Gamaliel Ble, Victor Castellanos, and Manuel Falconi. On the coexisting dynamics in the alternate iteration of two logistic maps. *Dynamical Systems*, 2011. URL <https://doi.org/10.1080/14689367.2011.554383>.
- Michael Bosse and Robert Zlot. Map matching and data association for large-scale two-dimensional laser scan-based slam. *The International Journal of Robotics Research*, 2008. URL <https://doi.org/10.1177/0278364908091366>.
- Russell Buchanan, Lorenz Wellhausen, Marko Bjelonic, Tirthankar Bandyopadhyay, Navinda Kottege, and Marco Hutter. Perceptive whole-body planning for multilegged robots in confined spaces. *Journal of Field Robotics*, 2021. URL <https://doi.org/10.1002/rob.21974>.
- Hsu-Yang Chang, C. S. George Lee, Yu Hu, and Yung-Hsiang Lu. Multi-robot slam with topological/metric maps. *Robotics literature*, 2007. URL <https://doi.org/10.1109/iro.2007.4399142>.
- Xiaotong Chen, Kaizhi Zheng, Zhen Zeng, Cameron Kisailus, Shreshtha Basu, James Cooney, Jana Pavlasek, and Odest Chadwicke Jenkins. Manipulation-oriented object perception in clutter through affordance coordinate frames. *arXiv*, 2020. URL <http://arxiv.org/abs/2010.08202v4>.
- Zhihong Chen, Junqiao Zhao, Tiantian Feng, Chen Ye, and Lu Xiong. Robust loop closure selection based on inter-robot and intra-robot consistency for multi-robot map fusion. *Remote Sensing*, 2023. URL <https://doi.org/10.3390/rs15112796>.
- S.H. Cheong, J.H. Lee, and C.H. Kim. A new concept of safety affordance map for robots object manipulation. *2018 27th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, 2018. URL <https://doi.org/10.1109/roman.2018.8525627>.
- Jiadi Cui and Soren Schwertfeger. Cp + : Camera poses augmentation with large-scale lidar maps. *2022 IEEE International Conference on Real-time Computing and Robotics (RCAR)*, 2022. URL <https://doi.org/10.1109/rcar54675.2022.9872176>.
- R. Detry, D. Kraft, O. Kroemer, L. Bodenhagen, J. Peters, N. Kruger, and J. Piater. Learning grasp affordance densities. *Paladyn, Journal of Behavioral Robotics*, 2011. URL <https://doi.org/10.2478/s13230-011-0012-x>.
- Kevin Doherty, Jinkun Wang, and Brendan Englot. Probabilistic map fusion for fast, incremental occupancy mapping with 3d hilbert maps. *2016 IEEE International Conference on Robotics and Automation (ICRA)*, 2016. URL <https://doi.org/10.1109/icra.2016.7487233>.
- Jun Dong and Jeffrey C. Trinkle. Orientation-based reachability map for robot base placement. *Robotics literature*, 2015. URL <https://doi.org/10.1109/iro.2015.7353564>.
- Mehmet R. Dougar. Affordances as a framework for robot control. *Robotics literature*, 2007. URL <https://openalex.org/W2159830351>.
- Gamal Elghazaly, Raphael Frank, Scott J. Harvey, and Stefan Saffko. High-definition maps: Comprehensive survey, challenges, and future perspectives. *IEEE Open Journal of Intelligent Transportation Systems*, 2023. URL <https://doi.org/10.1109/ojits.2023.3295502>.
- Lars-Peter Ellekilde, Shoudong Huang, Jaime Valls Miro, and Gamini Dissanayake. Dense 3d map construction for indoor search and rescue. *Journal of Field Robotics*, 2007. URL <https://doi.org/10.1002/rob.20173>.
- Parker Ewen, Adam Li, Yuxin Chen, Steven Hong, and Ram Vasudevan. These maps are made for walking: Real-time terrain property estimation for mobile robots. *IEEE Robotics and Automation Letters*, 2022. URL <https://doi.org/10.1109/lra.2022.3180439>.

- Amir Feder, Nadav Oved, Uri Shalit, and Roi Reichart. Causalm: Causal model explanation through counterfactual language models. *Computational Linguistics*, 2021. URL https://doi.org/10.1162/coli_a_00404.
- Lindner Felix and Eschenbach Carola. Affordances and affordance space: A conceptual framework for application in social robotics. *Frontiers in Artificial Intelligence and Applications Social Robots and the Future of Social Relations*, 2014. URL <https://doi.org/10.3233/978-1-61499-480-0-35>.
- Ross Finman, Thomas Whelan, Michael Kaess, and John J. Leonard. Efficient incremental map segmentation in dense rgb-d maps. *2014 IEEE International Conference on Robotics and Automation (ICRA)*, 2014. URL <https://doi.org/10.1109/icra.2014.6907666>.
- Farzin Foroughi, Jikai Wang, Alireza Nemati, Zonghai Chen, and Haoyuan Pei. Mapsegnet: A fully automated model based on the encoder-decoder architecture for indoor map segmentation. *IEEE Access*, 2021. URL <https://doi.org/10.1109/access.2021.3097893>.
- Avihai Giuli, Rotem Atari, and Avishai Sintov. Oracle-grasp: zero-shot affordance-aligned robotic grasping using large multimodal models. *Intelligent Service Robotics*, 2026. URL <https://doi.org/10.1007/s11370-026-00707-4>.
- Alex Guazzelli, Mihail Bota, Fernando J. Corbacho, and Michael A. Arbib. Affordances, motivations, and the world graph theory. *Adaptive Behavior*, 1998. URL <https://doi.org/10.1177/105971239800600305>.
- Dirk Hahnel, Rudolph Triebel, Wolfram Burgard, and Sebastian Thrun. Map building with mobile robots in dynamic environments. *Robotics literature*, 2004. URL <https://doi.org/10.1109/robot.2003.1241816>.
- Ji-Hyeong Han, Seungjae Lee, and Jong-Hwan Kim. Behavior hierarchy-based affordance map for recognition of human intention and its application to human-robot interaction. *IEEE Transactions on Human-Machine Systems*, 2016. URL <https://doi.org/10.1109/thms.2016.2558539>.
- Xuming He, Richard S. Zemel, and Volodymyr Mnih. Topological map learning from outdoor image sequences. *Journal of Field Robotics*, 2006. URL <https://doi.org/10.1002/rob.20170>.
- P. Hebert, Stephane Betge-Brezetz, and Raja Chatila. Decoupling odometry and exteroceptive perception in building a global world map of a mobile robot: the use of local maps. *Robotics literature*, 2002. URL <https://doi.org/10.1109/robot.1996.503865>.
- Elwan Hery, Philippe Xu, and Philippe Bonnifait. Consistent decentralized cooperative localization for autonomous vehicles using lidar, gnss, and hd maps. *Journal of Field Robotics*, 2021. URL <https://doi.org/10.1002/rob.22004>.
- Jiri Horner. Map-merging for multi-robot system. *Digital Repository (National Repository of Grey Literature)*, 2016. URL <https://openalex.org/W3005642736>.
- Chenguang Huang, Oier Mees, Andy Zeng, and Wolfram Burgard. Visual language maps for robot navigation. *arXiv*, 2022. URL <http://arxiv.org/abs/2210.05714v4>.
- Guanyu Huang and Roger K. Moore. Freedom comes at a cost?: An exploratory study on affordances’ impact on users’ perception of a social robot. *Frontiers in Robotics and AI*, 2024. URL <https://doi.org/10.3389/frobt.2024.1288818>.
- Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. Voxposer: Composable 3d value maps for robotic manipulation with language models. *arXiv (Cornell University)*, 2023. URL <https://doi.org/10.48550/arxiv.2307.05973>.
- Ander Iriondo, Elena Lazkano, and Ander Ansuategi. Affordance-based grasping point detection using graph convolutional networks for industrial bin-picking applications. *Sensors*, 2021. URL <https://doi.org/10.3390/s21030816>.

- Lorenzo Jamone, Emre Ugur, Angelo Cangelosi, Luciano Fadiga, Alexandre Bernardino, Justus Piater, and Jose Santos-Victor. Affordances in psychology, neuroscience, and robotics: A survey. *IEEE Transactions on Cognitive and Developmental Systems*, 2016. URL <https://doi.org/10.1109/tcds.2016.2594134>.
- Devendra K. Jangid, Neal R. Brodnik, McLean P. Echlin, Chandrakanth Gudavalli, Connor Levenson, Tresa M. Pollock, Samantha Daly, and B.S. Manjunath. Q-rbsa: high-resolution 3d ebsd map generation using an efficient quaternion transformer network. *npj Computational Materials*, 2024. URL <https://doi.org/10.1038/s41524-024-01209-6>.
- Peng Jiang, Nalin Bendapudi, Srikanth Saripalli, and Gaurav Pandey. 3dgs-loc: 3d gaussian splatting for map representation and visual localization. *Journal of Autonomous Vehicles and Systems*, 2025. URL <https://doi.org/10.1115/1.4069346>.
- Sing Bin Kang and Katsushi Ikeuchi. Robot task programming by human demonstration: mapping human grasps to manipulator grasps. *Robotics literature*, 2002. URL <https://doi.org/10.1109/iros.1994.407404>.
- Dimitrios Kanoulas, Jinoh Lee, Darwin G. Caldwell, and Nikos G. Tsagarakis. Visual grasp affordance localization in point clouds using curved contact patches. *International Journal of Humanoid Robotics*, 2017. URL <https://doi.org/10.1142/s0219843616500286>.
- Raj Surya Rajendran Kathirvel, Zach A Chavis, Stephen J. Guy, and Karthik Desingh. Sent map – semantically enhanced topological maps with foundation models. *arXiv*, 2025. URL <http://arxiv.org/abs/2511.03165v1>.
- Yuki Katsumata, Akinori Kanechika, Akira Taniguchi, Lotfi El Hafi, Yoshinobu Hagiwara, and Tadahiro Taniguchi. Map completion from partial observation using the global structure of multiple environmental maps. *Advanced Robotics*, 2022. URL <https://doi.org/10.1080/01691864.2022.2029762>.
- Kapil D. Katyal, Katie M. Popek, Chris Paxton, Phil Burlina, and Gregory D. Hager. Uncertainty-aware occupancy map prediction using generative networks for robot navigation. *Robotics literature*, 2019. URL <https://doi.org/10.1109/icra.2019.8793500>.
- Shafi Ullah Khan, Carla E. Garcia, Taewoong Hwang, and Insoo Koo. Radio environment map construction based on privacy-centric federated learning. *IEEE Access*, 2024. URL <https://doi.org/10.1109/access.2024.3367589>.
- David Inkyu Kim and Gaurav S. Sukhatme. Interactive affordance map building for a robotic task. *Robotics literature*, 2015. URL <https://doi.org/10.1109/iros.2015.7354029>.
- Doyeon Kim and Heoncheol Lee. Efficient 3d map merging with feature matching of projected maps in multi-robot slam. *The Journal of Korean Institute of Information Technology*, 2024. URL <https://doi.org/10.14801/jkiit.2024.22.5.71>.
- P. V. V. Kishore, D. Anil Kumar, Rama Chaithanya Tanguturi, K. Srinivasarao, P. Praveen Kumar, and D. Srihari. Joint motion affinity maps (jmam) and their impact on deep learning models for 3d sign language recognition. *IEEE Access*, 2024. URL <https://doi.org/10.1109/access.2024.3354775>.
- Kurt Konolige, D. Fox, Benson Limketkai, Jonathan Ko, and Benjamin Stewart. Map merging for distributed robot navigation. *Robotics literature*, 2004. URL <https://doi.org/10.1109/iros.2003.1250630>.
- Hema Swetha Koppula, Rudhir Gupta, and Ashutosh Saxena. Learning human activities and object affordances from rgb-d videos. *The International Journal of Robotics Research*, 2013. URL <https://doi.org/10.1177/0278364913478446>.
- Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew M. Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces. *arXiv (Cornell University)*, 2021. URL <https://doi.org/10.48550/arxiv.2108.08481>.

- Clayton Kunz and Hanumant Singh. Map building fusing acoustic and visual information using autonomous underwater vehicles. *Journal of Field Robotics*, 2013. URL <https://doi.org/10.1002/rob.21473>.
- Ivan Kuric, Vladimir Bulej, Milan Saga, and Peter Pokorny. Development of simulation software for mobile robot path planning within multilayer map system based on metric and topological maps. *International Journal of Advanced Robotic Systems*, 2017. URL <https://doi.org/10.1177/1729881417743029>.
- Phone Thiha Kyaw, Aung Paing, Theint Theint Thu, Mohan Rajesh Elara, Anh Vu Le, and Prabakaran Veerajagadheswar. Coverage path planning for decomposition reconfigurable grid-maps using deep reinforcement learning based travelling salesman problem. *IEEE Access*, 2020. URL <https://doi.org/10.1109/access.2020.3045027>.
- H.C. Lee, Byung-Hyuk Roh, and B.H. Lee. Multi-hypothesis map merging with sinogram-based pso for multi-robot systems. *Electronics Letters*, 2016. URL <https://doi.org/10.1049/el.2016.1041>.
- Heon-Cheol Lee, Seung Hwan Lee, Myoung Hwan Choi, and Beom-Hee Lee. Probabilistic map merging for multi-robot rbpf-slam with unknown initial poses. *Robotica*, 2011. URL <https://doi.org/10.1017/s026357471100049x>.
- Heoncheol Lee. Tomographic feature-based map merging for multi-robot systems. *Electronics*, 2020. URL <https://doi.org/10.3390/electronics9010107>.
- Hyeon-Seung Lee, Sung-Jun Kim, and Chan-Soo Kim. Real-time lidar localization on 3d gaussian splatting maps using hash-map search. *Journal of Digital Contents Society*, 2025. URL <https://doi.org/10.9728/dcs.2025.26.11.3281>.
- Sibaek Lee, Hyeonwoo Yu, Giseop Kim, and Sunwook Choi. Lamp: Implicit language map for robot navigation. *arXiv*, 2026. URL <http://arxiv.org/abs/2602.11862v1>.
- Filip Lefebvre, Dirk Lauwaet, Niels Souverijns, and Nele Veldeman. Urban heat maps in support of eu adaptation policy (u-map). *Robotics literature*, 2026. URL <https://doi.org/10.5194/egusphere-egu23-7081>.
- Feng Li, Young-Chul Kim, Ziang Lyu, and Han Zhang. Research on path planning for robot based on improved design of non-standard environment map with ant colony algorithm. *IEEE Access*, 2023a. URL <https://doi.org/10.1109/access.2023.3312940>.
- Feng Li, Young-Chul Kim, and Boyin Xu. Non-standard map robot path planning approach based on ant colony algorithms. *Sensors*, 2023b. URL <https://doi.org/10.3390/s23177502>.
- Xueting Li, Sifei Liu, Kihwan Kim, Xiaolong Wang, Ming-Hsuan Yang, and Jan Kautz. Putting humans in a scene: Learning affordance in 3d indoor environments. *arXiv*, 2019. URL <http://arxiv.org/abs/1903.05690v2>.
- Zhongqing Li, Xiong Liu, Huairui Wang, Song Jian, Fuxiang Xie, and Kai Wang. Research on robot path planning based on point cloud map in orchard environment. *IEEE Access*, 2024. URL <https://doi.org/10.1109/access.2024.3387539>.
- Yen-Chen Lin, Pete Florence, Andy Zeng, Jonathan T. Barron, Yilun Du, Wei-Chiu Ma, Anthony Simeonov, Alberto Rodriguez Garcia, and Phillip Isola. Mira: Mental imagery for robotic affordances. *arXiv (Cornell University)*, 2022. URL <https://doi.org/10.48550/arxiv.2212.06088>.
- Jiayang Liu, Xieyuanli Chen, Junhao Xiao, Sichao Lin, Zhiqiang Zheng, and Huimin Lu. Hybrid map-based path planning for robot navigation in unstructured environments. *Robotics literature*, 2023. URL <https://doi.org/10.1109/iros55552.2023.10341666>.
- Yifeng Liu and Jing Tian. Probabilistic attention map: A probabilistic attention mechanism for convolutional neural networks. *Sensors*, 2024. URL <https://doi.org/10.3390/s24248187>.

- Dong Ma, Teng Ma, Ye Li, Qiang Zhang, Yu Ling, and Yulei Liao. Underwater multizonotope terrain-aided navigation method with coarse map based on set-membership filter. *Journal of Field Robotics*, 2025. URL <https://doi.org/10.1002/rob.22407>.
- Jeffrey Mahler, Jacky Liang, Sherdil Niyaz, Michael Laskey, Richard Doan, Xinyu Liu, Juan Pablo Aparicio, and Ken Goldberg. Dex-net 2.0: Deep learning to plan robust grasps with synthetic point clouds and analytic grasp metrics. *Robotics literature*, 2017. URL <https://doi.org/10.15607/rss.2017.xiii.058>.
- Lucas Manuelli, Wei Gao, Pete Florence, and Russ Tedrake. kpm: Keypoint affordances for category-level robotic manipulation. *arXiv (Cornell University)*, 2019. URL <https://doi.org/10.48550/arxiv.1903.06684>.
- Tim K. Marks, Andrew Howard, Max Bajracharya, Garrison W. Cottrell, and Larry H. Matthies. Gamma-slam: Visual slam in unstructured environments using variance grid maps. *Journal of Field Robotics*, 2009. URL <https://doi.org/10.1002/rob.20273>.
- Yash Mehan, Kumaraditya Gupta, Rohit Jayanti, Anirudh Govil, Sourav Garg, and Madhava Krishna. Questmaps: Queryable semantic topological maps for 3d scene understanding. *arXiv*, 2024. URL <http://arxiv.org/abs/2404.06442v2>.
- Marcio Mendonca, Henrique Seiti Kondo, Lucas Botoni de Souza, Rodrigo Henrique Cunha Palacios, and Joao Paulo Lima Silva de Almeida. Semi-unknown environments exploration inspired by swarm robotics using fuzzy cognitive maps. *Robotics literature*, 2019. URL <https://doi.org/10.1109/fuzz-ieee.2019.8858847>.
- Marcio Mendonca, Rodrigo Henrique Cunha Palacios, Elpiniki I. Papageorgiou, and Lucas Botoni de Souza. Multi-robot exploration using dynamic fuzzy cognitive maps and ant colony optimization. *Robotics literature*, 2020. URL <https://doi.org/10.1109/fuzz48607.2020.9177814>.
- Simone Mentasti, Yusuf Can Simsek, and Matteo Matteucci. Traffic lights detection and tracking for hd map creation. *Frontiers in Robotics and AI*, 2023. URL <https://doi.org/10.3389/frobt.2023.1065394>.
- Malcolm Mielle, Martin Magnusson, and Achim J. Lilienthal. A method to segment maps from different modalities using free space layout maoris: Map of ripples segmentation. *2018 IEEE International Conference on Robotics and Automation (ICRA)*, 2018. URL <https://doi.org/10.1109/icra.2018.8461128>.
- Huaqing Min, Chang'an Yi, Ronghua Luo, Jinhui Zhu, and Sheng Bi. Affordance research in developmental robotics: A survey. *IEEE Transactions on Cognitive and Developmental Systems*, 2016. URL <https://doi.org/10.1109/tcds.2016.2614992>.
- Hans Moravec and Alberto Elfes. High resolution maps from wide angle sonar. *Robotics literature*, 2005. URL <https://doi.org/10.1109/robot.1985.1087316>.
- Masaki Murooka, Iori Kumagai, Mitsuharu Morisawa, and Fumio Kanehiro. Learning differentiable reachability maps for optimization-based humanoid motion generation. *arXiv*, 2025. URL <http://arxiv.org/abs/2508.11275v1>.
- Kevin P. Murphy. Bayesian map learning in dynamic environments. *Robotics literature*, 1999. URL <https://openalex.org/W2112793821>.
- Soroush Nasiriany, Sean Kirmani, Tianli Ding, Laura Smith, Yuke Zhu, Danny Driess, Dorsa Sadigh, and Ted Xiao. Rt-affordance: Affordances are versatile intermediate representations for robot manipulation. *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. URL <https://doi.org/10.1109/icra55743.2025.11127525>.
- J.I. Nieto, J.E. Guivant, and E.M. Nebot. The hybrid metric maps (hymms): a novel map representation for denseslam. *IEEE International Conference on Robotics and Automation, 2004. Proceedings. ICRA '04. 2004*, 2004. URL <https://doi.org/10.1109/robot.2004.1307181>.

- Negin Nikafrooz and Alexander Leonessa. A single-actuated, cable-driven, and self-contained robotic hand designed for adaptive grasps. *Robotics*, 2021. URL <https://doi.org/10.3390/robotics10040109>.
- Simon T O’Callaghan and Fabio T Ramos. Gaussian process occupancy maps. *The International Journal of Robotics Research*, 2012. URL <https://doi.org/10.1177/0278364911421039>.
- Giuseppe Oriolo, Giovanni Ulivi, and Marilena Vendittelli. Real-time map building and navigation for autonomous robots in unknown environments. *IEEE Transactions on Systems Man and Cybernetics Part B (Cybernetics)*, 1998. URL <https://doi.org/10.1109/3477.678626>.
- S.T. Pfister, Stergios I. Roumeliotis, and Joel W. Burdick. Weighted line fitting algorithms for mobile robot map building and efficient data representation. *Robotics literature*, 2004. URL <https://doi.org/10.1109/robot.2003.1241772>.
- Lasitha Piyathilaka and Sarath Kodagoda. Active visual object search using affordance-map in real world: A human-centric approach. *2014 13th International Conference on Control Automation Robotics and Vision (ICARCV)*, 2014. URL <https://doi.org/10.1109/icarcv.2014.7064525>.
- Lasitha Piyathilaka and Sarath Kodagoda. Affordance-map: Mapping human context in 3d scenes using cost-sensitive svm and virtual human models. *2015 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, 2015. URL <https://doi.org/10.1109/robio.2015.7419073>.
- William Prew, Toby P. Breckon, Magnus Bordewich, and Ulrik Beierholm. Evaluating gaussian grasp maps for generative grasping models. *arXiv*, 2022. URL <http://arxiv.org/abs/2206.00432v1>.
- Xianyu Qi, Wei Wang, Yuan Mei, Yuliang Wang, Mingbo Li, Xue Lin, and Yingpin Sun. Building semantic grid maps for domestic robot navigation. *International Journal of Advanced Robotic Systems*, 2020. URL <https://doi.org/10.1177/1729881419900066>.
- Weibiao Qiao and Zhe Yang. Modified dolphin swarm algorithm based on chaotic maps for solving high-dimensional function optimization problems. *IEEE Access*, 2019. URL <https://doi.org/10.1109/access.2019.2931910>.
- Peteris Racinskis, Janis Arents, and Modris Greitans. Constructing maps for autonomous robotics: An introductory conceptual overview. *Electronics*, 2023. URL <https://doi.org/10.3390/electronics12132925>.
- V. S. Rajashekhar and Gowdham Prabhakar. Developments and trend maps for soft continuum manipulators and soft snake robots. *Foundations and Trends in Robotics*, 2024. URL <https://doi.org/10.1561/23000000082>.
- Francesco Riccio, Roberto Capobianco, and Daniele Nardi. Learning human-robot handovers through -stam: Policy improvement with spatio-temporal affordance maps. *2016 IEEE-RAS 16th International Conference on Humanoid Robots (Humanoids)*, 2016. URL <https://doi.org/10.1109/humanoids.2016.7803373>.
- Barry Ridge, Danijel Skocaj, and Ales Leonardis. Self-supervised cross-modal online learning of basic object affordances for developmental robotic systems. *Robotics literature*, 2010. URL <https://doi.org/10.1109/robot.2010.5509544>.
- Jose Luis Susa Rincon and Stefano Carpin. Map merging of oriented topological semantic maps. *2019 International Symposium on Multi-Robot and Multi-Agent Systems (MRS)*, 2019. URL <https://doi.org/10.1109/mrs.2019.8901093>.
- Erol Sahin, Maya Cakmak, Mehmet R. Dogar, Emre Ugur, and Gokturk Ucoluk. To afford or not to afford: A new formalization of affordances toward affordance-based robot control. *Adaptive Behavior*, 2007. URL <https://doi.org/10.1177/1059712307084689>.

- Manish Saini, Melvin Paul Jacob, Minh Nguyen, and Nico Hochgeschwender. Planning robot placement for object grasping. *arXiv*, 2024. URL <http://arxiv.org/abs/2405.16692v1>.
- Luis Santos, Andre Aguiar, Filipe Neves dos Santos, Antonio Valente, and Marcelo R. Petry. Occupancy grid and topological maps extraction from satellite images for path planning in agricultural robots. *Robotics*, 2020. URL <https://doi.org/10.3390/robotics9040077>.
- Yuming Shi and Qiuju Xing. Dense distribution of chaotic maps in continuous map spaces. *Dynamical Systems*, 2011. URL <https://doi.org/10.1080/14689367.2011.627836>.
- Ola Shorinwa, Jiankai Sun, Mac Schwager, and Anirudha Majumdar. Siren: Semantic, initialization-free registration of multi-robot gaussian splatting maps. *arXiv*, 2025. URL <http://arxiv.org/abs/2502.06519v1>.
- Rakesh Shrestha, Fei-Peng Tian, Wei Feng, Ping Tan, and Richard Vaughan. Learned map prediction for enhanced mobile robot exploration. *Robotics literature*, 2019. URL <https://doi.org/10.1109/icra.2019.8793769>.
- Jaspreet Singh, Thomas Litfin, Jaswinder Singh, Kuldip Paliwal, and Yaoqi Zhou. Spot-contact-lm: improving single-sequence-based prediction of protein contact map using a transformer language model. *Bioinformatics*, 2022. URL <https://doi.org/10.1093/bioinformatics/btac053>.
- Marjorie Skubic, Sam Blisard, Andy Carle, and Pascal Matsakis. Hand-drawn maps for robot navigation. *Robotics literature*, 2002. URL <https://openalex.org/W178227448>.
- Jie Sun, Joshua L. Moore, Aaron Bobick, and James M. Rehg. Learning visual object categories for robot affordance prediction. *The International Journal of Robotics Research*, 2010. URL <https://doi.org/10.1177/0278364909356602>.
- Francesco Taioli, Federico Cunico, Federico Girella, Riccardo Bologna, Alessandro Farinelli, and Marco Cristani. Language-enhanced rnn-map: Querying renderable neural radiance field maps with natural language. *2023 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2023. URL <https://doi.org/10.1109/iccvw60793.2023.00504>.
- Kentaro Takemura, Ato Araki, Junichi Ido, Yoshio Matsumoto, Jun Takamatsu, and Tsukasa Ogasawara. Generating individual maps from universal map for heterogeneous mobile robots. *2010 IEEE International Conference on Robotics and Automation*, 2010. URL <https://doi.org/10.1109/robot.2010.5509418>.
- Ben Talbot, Feras Dayoub, Peter Corke, and Gordon Wyeth. Robot navigation in unseen spaces using an abstract map. *QUT ePrints (Queensland University of Technology)*, 2021. URL <https://openalex.org/W3099859956>.
- Tim Y. Tang, Daniele De Martini, Shangzhe Wu, and Paul Newman. Self-supervised learning for using overhead imagery as maps in outdoor range sensor localization. *The International Journal of Robotics Research*, 2021. URL <https://doi.org/10.1177/02783649211045736>.
- Sebastian Thrun, Jens-Steffen Gutmann, Dieter Fox, Wolfram Burgard, and Benjamin Kuipers. Integrating topological and metric maps for mobile robot navigation: a statistical approach. *National Conference on Artificial Intelligence*, 1998. URL <https://openalex.org/W2153852998>.
- Nagasaka Tomomi and Tanaka Kanji. Dictionary-based map compression for sparse feature maps. *2011 IEEE International Conference on Robotics and Automation*, 2011. URL <https://doi.org/10.1109/icra.2011.5979638>.
- Edmond Tong, Advait Balaji, Anthony Pipari, Stanley Lewis, Zhen Zeng, and Odest Chadwicke Jenkins. Oval-grasp: Open-vocabulary affordance localization for task oriented grasping. *arXiv*, 2025. URL <http://arxiv.org/abs/2511.20841v1>.

- Jose Miguel Torres-Camara, Felix Escalona, Francisco Gomez-Donoso, and Miguel Cazorla. Map slammer: Densifying scattered kslam 3d maps with estimated depth. *Advances in Intelligent Systems and Computing Robot 2019: Fourth Iberian Robotics Conference*, 2020. URL https://doi.org/10.1007/978-3-030-36150-1_46.
- Emre Ugur and Erol Sahin. Traversability: A case study for learning and perceiving affordances in robots. *Adaptive Behavior*, 2010. URL <https://doi.org/10.1177/1059712310370625>.
- Peter Veres. MI and statistics-driven route planning: Effective solutions without maps. *Logistics*, 2025. URL <https://doi.org/10.3390/logistics9030124>.
- Panagiotis Vlantis, Constantinos Vrohidis, Charalampos P. Bechlioulis, and Kostas J. Kyriakopoulos. Robot navigation in complex workspaces using harmonic maps. *Robotics literature*, 2018. URL <https://doi.org/10.1109/icra.2018.8460695>.
- Maozhen Wang, Rui Luo, Aykut Ozgun Onol, and Taskn Padr. Affordance-based mobile robot navigation among movable obstacles. *Robotics literature*, 2020a. URL <https://doi.org/10.1109/iros45743.2020.9341337>.
- Qianlong Wang, Tianxi Ji, Yifan Guo, Lixing Yu, Xuhui Chen, and Pan Li. Trafficchain: A blockchain-based secure and privacy-preserving traffic map. *IEEE Access*, 2020b. URL <https://doi.org/10.1109/access.2020.2980298>.
- Ruixiang Wang, Huayi Zhou, Xinyue Yao, Guiliang Liu, and Kui Jia. Gat-grasp: Gesture-driven affordance transfer for task-aware robotic grasping. *arXiv*, 2025a. URL <http://arxiv.org/abs/2503.06227v1>.
- Shengfan Wang, Xin Jiang, Jie Zhao, Xiaoman Wang, Weiguo Zhou, and Yunhui Liu. Efficient fully convolution neural network for generating pixel wise robotic grasps with high resolution images. *Robotics literature*, 2019. URL <https://doi.org/10.1109/robio49542.2019.8961711>.
- Yingyu Wang, Liang Zhao, and Shoudong Huang. Occupancy-slam: An efficient and robust algorithm for simultaneously optimizing robot poses and occupancy map. *IEEE Transactions on Robotics*, 2025b. URL <https://doi.org/10.1109/tro.2025.3578227>.
- Arie Wisianto, Hidayatus Saniya, and Oki Gumilar. Integrating pipeline data management application and google maps dataset on web based gis application using open source technology sharp map and open layers. *2010 8th International Pipeline Conference, Volume 3*, 2010. URL <https://doi.org/10.1115/ipc2010-31438>.
- Heng Wu, Yanjie Liu, Chao Wang, and Yanlong Wei. An effective 3d instance map reconstruction method based on rgb-d images for indoor scene. *Remote Sensing*, 2025. URL <https://doi.org/10.3390/rs17010139>.
- Oliver Wulf, Andreas Nuchter, Joachim Hertzberg, and Bernardo Wagner. Benchmarking urban six-degree-of-freedom simultaneous localization and mapping. *Journal of Field Robotics*, 2008. URL <https://doi.org/10.1002/rob.20234>.
- Pengwei Xie, Siang Chen, Qianrun Chen, Wei Tang, Dingchang Hu, Yixiang Dai, Rui Chen, and Guijin Wang. Gap-rl: Grasps as points for rl towards dynamic object grasping. *arXiv*, 2024. URL <http://arxiv.org/abs/2410.03509v1>.
- Wenting Xu, Viorela Ila, Luping Zhou, and Craig T. Jin. Tb-hsu: Hierarchical 3d scene understanding with contextual affordances. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025. URL <https://doi.org/10.1609/aaai.v39i9.32969>.
- Guanghui Xue, Ruixue Li, Shuang Liu, and Jinbo Wei. Research on underground coal mine map construction method based on lego-loam improved algorithm. *Energies*, 2022. URL <https://doi.org/10.3390/en15176256>.

- Kai Yan, Lanyue Liang, Ziqiang Zheng, Guoqing Wang, and Yang Yang. Medium transmission map matters for learning to restore real-world underwater images. *Applied Sciences*, 2022. URL <https://doi.org/10.3390/app12115420>.
- Bincheng Yang and Hongwei Li. A visual attention model based on eye tracking in 3d scene maps. *ISPRS International Journal of Geo-Information*, 2021. URL <https://doi.org/10.3390/ijgi10100664>.
- Xintong Yang, Ze Ji, Jing Wu, and Yu-Kun Lai. Recent advances of deep robotic affordance learning: A reinforcement learning perspective. *IEEE Transactions on Cognitive and Developmental Systems*, 2023. URL <https://doi.org/10.1109/tcds.2023.3277288>.
- Shunyi Yao, Guangda Chen, Quecheng Qiu, Jun Ma, Xiaoping Chen, and Jianmin Ji. Crowd-aware robot navigation for pedestrians with multiple collision avoidance strategies via map-based deep reinforcement learning. *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021. URL <https://doi.org/10.1109/iro51168.2021.9636579>.
- Shuiwen Yu, Chunyun Fu, Amirali Khodadadian Gostar, and Minghui Hu. A review on map-merging methods for typical map types in multiple-ground-robot slam solutions. *Sensors*, 2020. URL <https://doi.org/10.3390/s20236988>.
- Wentao Yuan, Jiafei Duan, Valts Blukis, Wilbert Pumacay, Ranjay Krishna, Adithyavairavan Murali, Arsalan Mousavian, and Dieter Fox. Robopoint: A vision-language model for spatial affordance prediction for robotics. *arXiv*, 2024. URL <http://arxiv.org/abs/2406.10721v1>.
- P Zech, Simon Haller, Safoura Rezapour Lakani, Barry Ridge, Emre Ugur, and Justus Piater. Computational models of affordance in robotics: a taxonomy and systematic classification. *Adaptive Behavior*, 2017. URL <https://doi.org/10.1177/1059712317726357>.
- Andy Zeng, Shuran Song, Kuan-Ting Yu, Elliott Donlon, Francois R. Hogan, Maria Bauza, Daolin Ma, Orion Taylor, Melody Liu, Eudald Romo, Nima Fazeli, Ferran Alet, Nikhil Chavan-Dafle, Rachel Holladay, Isabella Morona, Prem Nair, Druck Green, Ian Taylor, Weber Liu, and Thomas Funkhouser. Robotic pick-and-place of novel objects in clutter with multi-affordance grasping and cross-domain image matching. *The International Journal of Robotics Research*, 2019. URL <https://doi.org/10.1177/0278364919868017>.
- Jingrui Zhang, Zhenhong Xu, Houde Liu, Xiaojun Zhu, and Bin Lan. An improved hybrid ant colony optimization and genetic algorithm for multi-map path planning of rescuing robots in mine disaster scenario. *Machines*, 2025a. URL <https://doi.org/10.3390/machines13060474>.
- Xiaoyu Zhang, Jinhu Dong, Yin Zhang, and Yun-Hui Liu. Ms-slam: Memory-efficient visual slam with sliding window map sparsification. *Journal of Field Robotics*, 2025b. URL <https://doi.org/10.1002/rob.22431>.
- Xuechao Zhang, Dong Wang, Sun Han, Weichuang Li, Bin Zhao, Zhigang Wang, Xiaoming Duan, Chongrong Fang, Xuelong Li, and Jianping He. Affordance-driven next-best-view planning for robotic grasping. *arXiv*, 2023. URL <http://arxiv.org/abs/2309.09556v2>.
- Ying Zhang, Guohui Tian, Xuyang Shao, Shaopeng Liu, Mengyang Zhang, and Peng Duan. Building metric-topological map to efficient object search for mobile robot. *IEEE Transactions on Industrial Electronics*, 2021. URL <https://doi.org/10.1109/tie.2021.3095812>.
- Chengxi Zhao, Linjiang Chen, Yu Che, Zhongfu Pang, Xiaofeng Wu, Yunxiang Lu, Honglai Liu, Graeme M. Day, and Andrew I. Cooper. Digital navigation of energy-structure-function maps for hydrogen-bonded porous molecular crystals. *Nature Communications*, 2021. URL <https://doi.org/10.1038/s41467-021-21091-w>.
- Liang Zhao, Yingyu Wang, and Shoudong Huang. Occupancy-slam: Simultaneously optimizing robot poses and continuous occupancy map. *arXiv*, 2024. URL <http://arxiv.org/abs/2405.10743v1>.
- Weiming Zhi, Lionel Ott, Ransalu Senanayake, and Fabio Ramos. Continuous occupancy map fusion with fast bayesian hilbert maps. *2019 International Conference on Robotics and Automation (ICRA)*, 2019. URL <https://doi.org/10.1109/icra.2019.8793508>.

1296 Liang Zhou, Kevin Smith, Joshua Tenenbaum, and Tobias Gerstenberg. Mental jenga: A counter-
1297 factual simulation model of causal judgments about physical support. *Robotics literature*, 2026.
1298 URL <https://doi.org/10.31234/osf.io/4a5uh>.
1299
1300 Zhiguo Zhou, Xin Feng, Shunfan Di, and Xuehua Zhou. A lidar mapping system for robot navigation
1301 in dynamic environments. *IEEE Transactions on Intelligent Vehicles*, 2023. URL <https://doi.org/10.1109/tiv.2023.3328013>.
1302
1303 Xingxing Zuo, Wenlong Ye, Yulin Yang, Renjie Zheng, Teresa Vidal-Calleja, Guoquan Huang, and
1304 Yong Liu. Multimodal localization: Stereo over lidar map. *Journal of Field Robotics*, 2020. URL
1305 <https://doi.org/10.1002/rob.21936>.
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